

Name _____ Date _____ Period _____

Comparing All Seven Parent Functions

Algebra II Honors | Unit 1 | Day 8 | TEK A2.2.A

Objective: I can compare all seven parent functions across their key attributes, classify each as even, odd, or neither using an algebraic test, and state end behavior in formal notation.

Vocabulary

Even: _____ **Odd:** _____ Most functions are _____.

Relative max / min — the largest or smallest output _____, which may exist even when no overall one does.

Part 1 — The Master Table

Complete every cell from your Day 4-7 work. This is your reference for the year.

Function	Domain	Range	Intercept	Asymptotes
$\sqrt{x}$	_____	_____	_____	_____
$ x $	_____	_____	_____	_____
x^3	_____	_____	_____	_____
$\sqrt[3]{x}$	_____	_____	_____	_____
$1/x$	_____	_____	_____	_____
b^x	_____	_____	_____	_____
log base b	_____	_____	_____	_____

Part 2 — Even, Odd, or Neither

Substitute $-x$ into each rule. Do not decide from the picture alone.

Function	$f(-x)$ simplifies to	Even, odd, or neither
$ x $	_____	_____
x^3	_____	_____
$\sqrt[3]{x}$	_____	_____
$1/x$	_____	_____
$\sqrt{x}$	_____	_____
2^x	_____	_____

$\sqrt{x}$ is "neither" for a different reason than 2^x . Explain the difference:

$\sqrt[3]{-x}$ is _____ for positive x , while 2^x is defined but _____.

Part 3 — Name That Function

1. Odd, no intercepts, two asymptotes. _____
2. Even, minimum 0, domain all reals. _____
3. Domain $(0, \infty)$, range all reals, through $(1, 0)$. _____

4. Odd, domain and range all reals, and steepens as x grows. _____

5. Items 4 and its partner $\sqrt[3]{x}$ are both odd with the same domain and range. What single piece of information separates them?

One point, such as _____

Part 4 — Extremes on an Interval

a. $f(x) = x^3$ on $[-2, 2]$: min _____ max _____ b. $f(x) = |x|$ on $[-3, 1]$: min _____
max _____

c. $f(x) = 1/x$ on $[1, 4]$: min _____ max _____ d. $f(x) = 1/x$ on $[-1, 1]$:

Item (d) is the interesting one. Why does a closed interval fail to guarantee a max and min here?

The function is _____ — there is a break at $x = 0$.

Part 5 — Going Further

1. Sort all seven by whether the domain is restricted. restricted: _____

2. Only two of the seven have a minimum. Name them and explain what they share.

$\sqrt{x}$ and $|x|$ — _____

3. Must an even function pass through the origin? Give an example or counterexample.

No — _____

Exit Ticket

Classify $f(x) = 1/x$ as even, odd, or neither, showing the algebra, and name one other parent function in the same class.

$f(-x) =$ _____ Another odd one: _____